

James Nithil V

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EXPERIENCE

Junior Software Developer

Aug 2025 to Present

Vaken Technology | Product: Sovablu

Trichy, India

- Engineered and scaled enterprise RESTful APIs using **Java 21, Spring Boot, and PostgreSQL**, implementing strict **Spring Transaction management and ACID principles** across critical mutation and confirmation workflows to maintain data consistency and prevent data loss under concurrent client traffic.
- Architected a hybrid persistence layer pairing **PostgreSQL** with **Amazon DynamoDB** for fast key value reads; restructured table designs and eliminated redundant columns to reduce I/O payload size while implementing **composite indexing and query optimization** to eliminate full row scans.
- Implemented an in memory **Redis caching tier with TTL expiration** for frequently queried datasets to reduce repeated database queries; applied **cache stampede prevention mechanisms** using mutex locks to safeguard the database from concurrent traffic spikes and improve API response times.
- Secured file and document storage using **Amazon S3 with expiring presigned URLs** and enforced **role based access control (RBAC)** across platform endpoints; automated Java compilation, testing, and containerized deployment to **Amazon ECS** via **Jenkins CI/CD** with **AWS Lambda** for event driven scheduled tasks.
- Contributed to **Agentic AI development** by building autonomous agents, defining modular skills, integrating **MCP tool access**, and deploying **RAG pipelines with guardrails**; optimized LLM inference costs and response efficiency through **dynamic model routing**, directing routine operations to lightweight models and reserving complex reasoning for advanced LLMs.

PROJECTS

KadalVazhi: Autonomous Maritime Intelligence Platform

2026

Java, Spring Boot, Python, LangGraph, Kafka, PostgreSQL, DynamoDB, gRPC

Production Microservices

- Architected distributed maritime engine with **Java** and **Spring Boot** for core backend security and **Python** for **LangGraph multi agent orchestration** and **Claude SDK model routing**.
- Engineered microservices communication layer utilizing **Spring Boot, REST APIs, GraphQL, and gRPC** for high throughput vessel telemetry, direct fish trade, and automated **emergency mechanic dispatch agents** for at sea boat repairs.
- Implemented **dual database architecture with PostgreSQL and DynamoDB**, backed by **Apache Kafka** event streaming to scale concurrent vessel telemetry and power dynamic **fisherman and boat crew job matchmaking**.
- Designed resilient **offline first synchronization engine**: logs vessel telemetry in local SQLite at sea and auto syncs to cloud PostgreSQL upon cellular reconnection, with a 3 attempt fallback loop for weather safety.
- Built comprehensive fish catch analytics engine aggregating historical catch data for data driven voyage decisions, supporting multilingual accessibility across **Tamil, English, and Malayalam via i18n**.

TECHNICAL SKILLS

Languages: Java (Core & 21), Python, SQL, C++, JavaScript

Backend & APIs: Spring Boot, FastAPI, REST APIs, GraphQL, gRPC, Microservices

Distributed Systems & Cloud: Apache Kafka, Distributed Systems, Redis Caching, Rate Limiting, Docker, Jenkins CI/CD, AWS (EC2, S3, Lambda, Bedrock)

Databases: PostgreSQL, DynamoDB, MySQL, Redis, SQLite

AI & Systems: LangGraph, Agentic AI, Multi Agent Orchestration, Claude SDK, LangSmith, i18n

ACHIEVEMENTS

- LeetCode Knight Badge:** Earned an official **Contest Rating of 1,994** (Top 2.73% worldwide); solved **700+ algorithmic problems** in **Java and Python** (Graphs, Trees, DP, BFS/DFS, Linked Lists).
- Smart India Hackathon (SIH) Team Lead:** Led a 5 member engineering team to build **FarmVista**, an AI precision agriculture platform integrating deep learning computer vision for automated crop disease diagnosis and soil telemetry.

EDUCATION

Knowledge Institute of Technology

Salem, India

B.Tech in Artificial Intelligence and Data Science | CGPA: 8.1

2021 to 2025

CERTIFICATIONS

Core Java & SQL (EBox) | MySQL Database (IBM) | Machine Learning (Simplilearn) | Deep Learning PyTorch (Guvi)